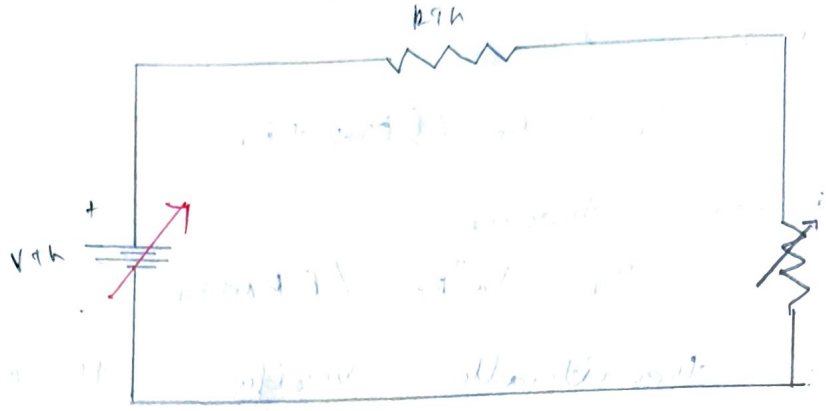
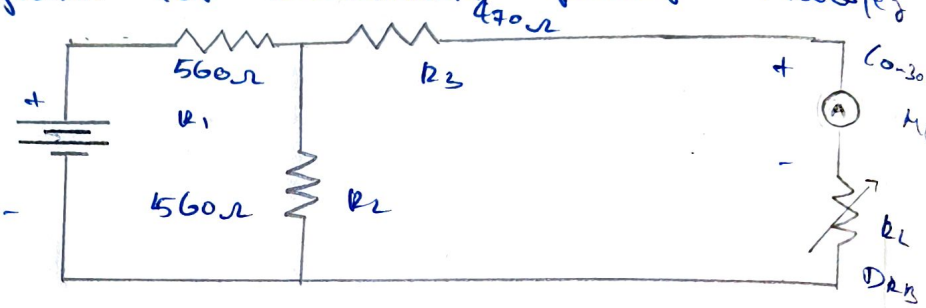
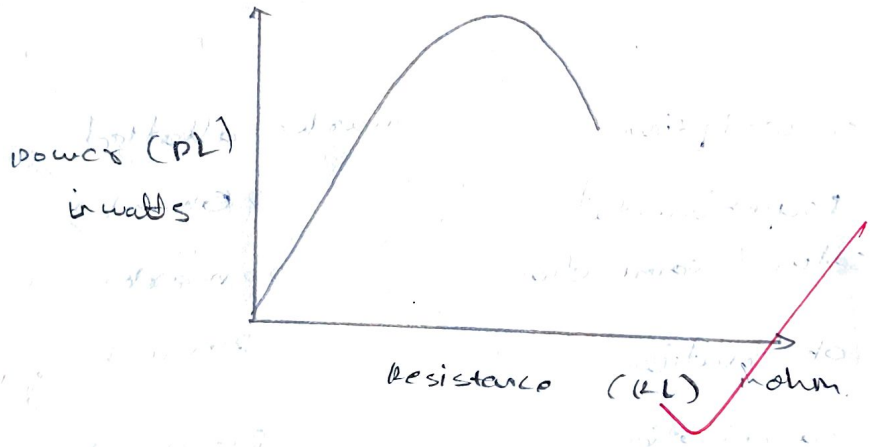


Circuit Diagram: For maximum power Transfer theorem:



model graph:



Exp.No: 6

DATE: 21/02/2026 verification of maximum power transfer theorem

Aim:

* To measure the power absorbed in a load and to verify that the power absorbed in a load is maximum only when load resistance is equal to the source resistance

Apparatus Required:

S.NO	Name of the apparatus	Range /Rating	Quantity
1	voltmeter	(0-15V) MC	1
2	Ammeter	(0-500mA) MC	1
3	Resistor	560 Ω 47Ω	2 1
4	RPS (DC supply)	15 V	1

Theory:

* The maximum power transfer states that "A load will receive maximum power from a linear bilateral DC network when its total resistive value is exactly equal to the thevenin resistance of the networks as seen by load".

Tabulation:

Table (1): For the oretical calculation : $R_{th} = 750\Omega$

S.NO	Load Resistance (R_L) in ohm	Load current (I_L) in mA	Load Voltage (V_L) in volts	Load power (P_L) in watts)
1.	750Ω	3.33A	2.492	8.39 mW

Table (2): For the practical calculation:

S.NO	Load Resistance (R_L) in ohm	Load current (I_L) in amps	Load Voltage (V_L) in volts	Load power (P_L) in watts
1.	330	4.9×10^{-3}	1.551	7.29×10^{-3}
2	470	4.20×10^{-3}	1.974	8.29×10^{-3}
3.	500	3.3×10^{-3}	2.64	8.71×10^{-3}
4.	1 k	2.9×10^{-3}	3.9	8.41×10^{-3}
5.	2 k	1.86×10^{-3}	3.72	6.92×10^{-3}
6.	3k	1.37×10^{-3}	4.11	5.63×10^{-3}

For the network in figure (a), maximum power will be delivered to the load when $R_L = R_{Th}$.

* In a simpler form the circuit may contain a voltage source V_s having internal resistance R_s and connected across a load R_L. To find power.

Procedure

* make connection as per the circuit diagram

* change the resistor R_L whose value close to R_{th}, measure the corresponding V_L, I_L

* plot a graph between R_L and P_L and find the R_L corresponding to maximum power transfer.

* verify the measure values of R_L at maximum power transfer as same as calculated and found graphically

Model calculation:

Theoretical calculation:

$V_s = 10V$

As VDR => $V_{Th} = V_s \times \frac{R_s}{R_s + R_L}$

$V_{th} = 10 \times \frac{560}{1120}$

$V_{Th} = 5V$

R_{th} = R_1 and R_2 is parallel

$= \frac{R_1 \times R_2}{R_1 + R_2} = \frac{560 \times 560}{560 + 560} = \frac{313600}{1120} = 280\Omega$

This is in series with R_3:

$R_{eq} = 280 + 470$

$= 750\Omega$

Having power transfers

$P = \frac{(V_{th})^2 R_L}{(R_{th} + R_L)}$

$R_L = R_{th}$

$P = \frac{V_{th}^2}{4R_2}$

$R_L = 750\Omega$

$P = \frac{5^2}{4 \times 750} = 8.39mW$

$I_L = \frac{V_{th}}{R_{th} + R_L} = 5/1500$

$I_L = 0.0033A$

$I_L = 3.33mA$

$V_L = I_L \times R_L$

$V_L = 3.33 \times 10^{-3} \times 750$

$V_L = 2.49V$

Sl.NO	Description	marks Alloted	marks obtained
1.	Experimental	20 marks	20
2.	Observation	20 marks	20
3.	Calculation	20 marks	18
4.	viva	20 marks	18
5	Record	20 marks	18
			94

Result:

* thus maximum power transfer theorem is verified practically and theoretically